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Teaching Interns

March 4, 2026
Date of Teaching

Mr. Sunny R. Fernandez
Teaching Internship Mentor

Grade 8/ WTh 10:00 AM - 11:50 AM
Class and Schedule

March 1, 2026
Date of Submission

March , 2026
Date of Approval

LESSON PLAN in Science 8
(1st Revision)

Content Standard	*The learners learn that:* ● kinetic energy is the energy of movement, and potential energy is stored energy. ● as an object falls from a height its energy is conserved because its potential energy is transformed to kinetic energy.
Performance Standard	By the end of the Quarter, the learners: ● link motion to kinetic energy and potential energy and explain transformations between them using everyday examples.
Learning Competencies	The learners: a. describe kinetic energy as the movement of objects or particles, and potential energy as energy stored due to the position of objects or particles; b. identify examples of everyday situations that demonstrate: ● kinetic energy being transformed to potential energy, and ● potential energy being transformed to kinetic energy; c. explain that the mechanical energy of an object is the sum of the kinetic energy and the potential energy available to do work; d. describe conservation of energy in everyday situations involving gravity, such as when objects fall;

I. Intended Learning Outcomes (Objectives)	
At the end of the lesson, the Grade 8 learners, with at least 80% accuracy, will be able to:	
Knowledge	Differentiate the fundamental physical properties of kinetic and potential energy in mechanical systems.
Skills	Model the conservation of total mechanical energy in an isolated system during motion.
Attitudes/Values	Acknowledge the importance of mechanical energy and work in our day to day lives.

II. Learning Content	
A. Topic (s)	**Lesson 4.3: Energy**
B. Key Concepts	• Energy is the capacity to do work. • Mechanical energy is the energy that is possessed by an object due to its motion or due to its position • Total mechanical energy is the sum of the kinetic and potential energy of a system • Kinetic energy is energy of motion, where $KE=\frac{1}{2}mv^2$ • Potential energy is stored energy due to an object's position or state, one specific type of potential energy is gravitational potential energy where PE=mgh • Energy in an isolated system is always conserved. Energy cannot be created, nor destroyed, only transformed from one form to another.
C. References *(Print, electronic, etc.)*	• Madriaga, E. A., Piloton-Narca, M., Mariano, J. J. M., Ledesma, J. R. C., & Apolinario-Ramos, E. G. (2025). *Science links 8: Worktext for scientific and technological literacy* (1st ed.). Rex Book Store.
D. Instructional Materials	• Laptop • Canva Presentation • Google Docs
E. Integration *(Values,*	**Values:** Accuracy and Resourcefulness **SDG 7:** (Affordable and Clean Energy)

SDG/ESD, GAD, etc.)	**SDG 9:** (Industry, Innovation, and Infrastructure)
F. Skills *(21st century skills)*	• Critical Thinking and Problem Solving • Communication Skills • Collaboration

III. Teaching and Learning Process (Experiences)

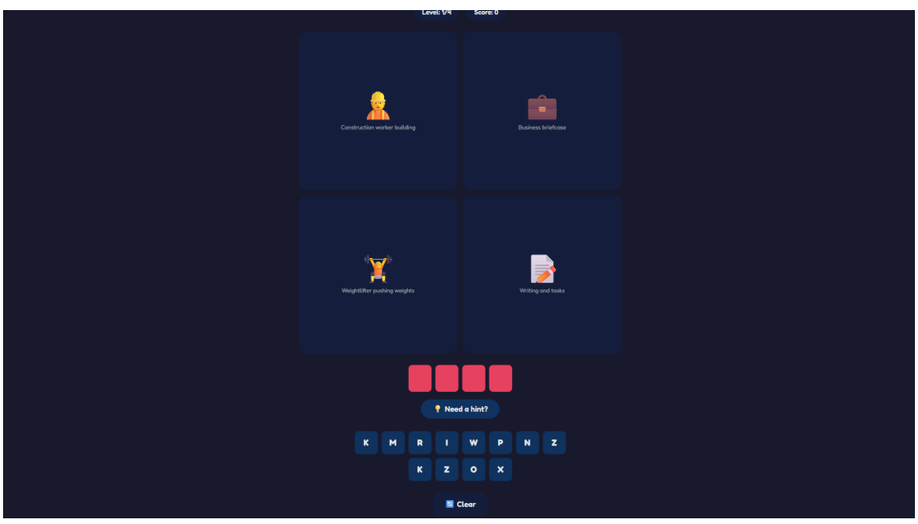
Model: 5E Instructional Model
Total time allotment: 90 minutes

PRELIMINARIES (5 minutes)

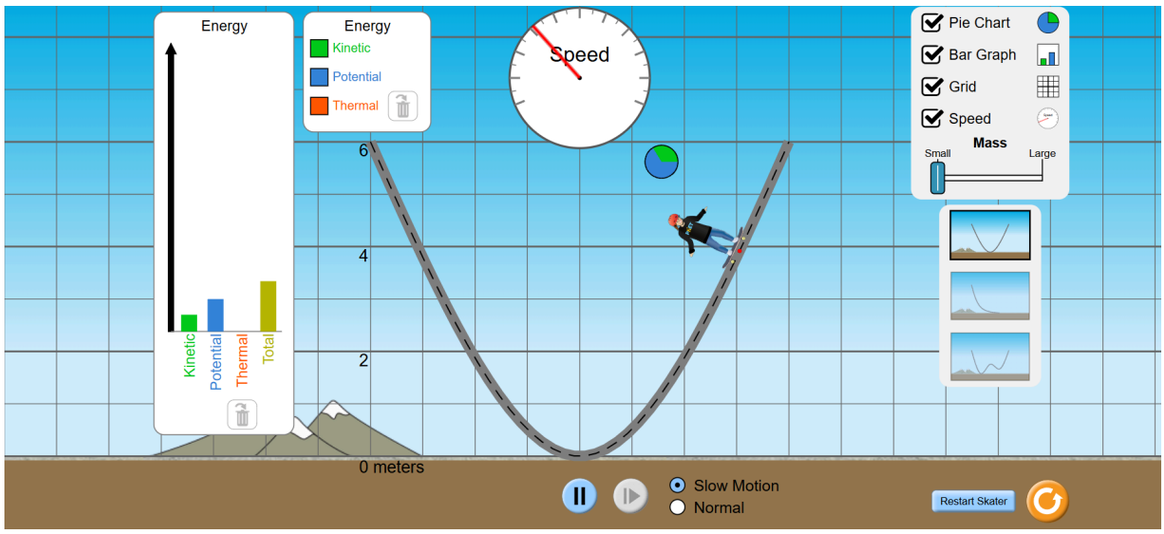
The teacher begins the lesson with a prayer, followed by attendance checking. The teacher will provide a brief recap of lesson 4.2: graphing uniform and non-uniform motion, before transitioning to engage.

A. ENGAGE (15 minutes)

The teacher will start an activity that will cultivate their knowledge about the topic by doing an activity called 4 emojis 1 word. This will also help the students to gain some vocabulary while giving a hint to what the topic is all about.



B. EXPLORE (15 minutes)



The teacher will distribute the direct link to the PhET Interactive Simulation: Energy Skate Park to the online class. While the teacher displays the simulation on screen, they will also instruct the students (who want to do it on their own) to activate the "Bar Graph" and "Speed" display options within the digital interface. The teacher will command the students to place the skater at the highest point of the U-shaped track and release it. During the simulation, the teacher will require the students to observe the bar graphs and record the exact track locations where kinetic energy and potential energy reach their absolute maximum and minimum values.

Additionally, the teacher will direct the students to monitor the "Total Energy" column throughout the continuous motion sequence. This is followed up with the following processing questions:

1. Identify the exact track location where the potential energy bar reached its absolute maximum. Where does the potential energy peak?

2. Identify the exact track location where the kinetic energy bar reached its absolute maximum. Where does the kinetic energy peak?

3. How does changing the mass affect the energy of the system?

4. Analyze the exact relationship between the potential and kinetic energy bars as the skater descends. What is the relationship of the two as one increases?

5. Observe the Total Energy bar. How does it change as the skater moves along the track?

6. If kinetic and potential energy continuously fluctuate, why does the Total Energy value remain static?

   Challenge: Is the answer for question 6 always true?

C. EXPLAIN (20 minutes)

The teacher will facilitate a discussion to synthesize the students' observations from the PhET simulation and formally introduce the key concepts and formulas.

The teacher will begin by revisiting the processing questions from the Explore phase to establish connections between student observations and scientific terminology:

1. Based on your observations, where was potential energy highest? Where was kinetic energy highest? What does this tell us about the relationship between height and these energy types?
2. You observed that when potential energy decreased, kinetic energy increased. What is this relationship called? How would you describe it in your own words?
3. What did you notice about the Total Energy bar throughout the simulation? Did it change? What does this suggest about the total amount of energy in the system?

The teacher will synthesize the answers of the students leading them to form these definitions:

- Mechanical Energy - the energy possessed by an object due to its motion or its position. It is the sum of kinetic and potential energy.

- Total Mechanical Energy - the sum of the kinetic energy and potential energy in a system, represented as ME = KE + PE.

- Kinetic Energy (KE) - the energy of motion. Any moving object possesses kinetic energy. It is dependent on the object's mass and velocity. The formula for kinetic energy is $KE = \frac{1}{2}mv^2$, where:

    - m = mass of the object (kg)
    - v = velocity of the object (m/s)

- Gravitational Potential Energy (PE) - stored energy due to an object's position or height above a reference point. It is dependent on the object's mass, the acceleration due to gravity, and its height. The formula for gravitational potential energy is PE = mgh, where:

    - m = mass of the object (kg)
    - g = acceleration due to gravity (9.8 m/s^2)
    - h = height above the reference point (m)

The teacher will use simple, everyday examples to illustrate these concepts, such as comparing a book on a low shelf versus a high shelf, or a moving bicycle versus a stationary bicycle.

The teacher will then introduce the Law of Conservation of Energy:

- Energy cannot be created nor destroyed; it can only be transformed from one form to another.

- In an isolated system (like the PhET simulation with no friction), the total mechanical energy remains constant. The fluctuations observed in the KE and PE bars demonstrate energy transformation, not energy loss.

$$KE_1 + PE_1 = KE_2 + PE_2$$

$$\frac{1}{2}mv^2_1 + mgh_1 = \frac{1}{2}mv^2_2 + mgh_2$$

Finally, the teacher will transition by posing a guiding question to lead into the next part of the Explain phase:

- If energy is being transformed between kinetic and potential forms, what is the mechanism that causes this transfer? What is actually being done to the object to change its energy? This leads us to the concept of Work.

The teacher will transition from the Law of Conservation of Energy by posing a guiding question: "We now know that energy transforms from potential to kinetic and back. But what do we call the actual process of energy moving from one form to another? What is the mechanism that transfers energy?"

The teacher will then introduce the concept of Work using its energy-based definition:

- Work is the process of transferring energy from one object to another or transforming energy from one form to another.
- When work is done on an object, energy is transferred to that object.
- When work is done by an object, energy is transferred away from that object.
- Work is not a form of energy itself; rather, it is the mechanism or the process by which energy is transferred or transformed.

The teacher will provide concrete examples that illustrate work as energy transfer without referencing force:

1. Lifting a book: When you lift a book from the floor to a shelf, you are doing work on the book. Before lifting, the book had low potential energy (low height). After lifting, the book has higher potential energy. Where did that additional potential energy come from? It came from you. You transferred energy from your body to the book. The process of lifting—the act of transferring that energy—is called work.

2. A falling ball: When a ball falls from a height, gravity does work on the ball. The ball loses potential energy and gains kinetic energy. The work done by gravity is the process that transforms potential energy into kinetic energy.

The teacher will emphasize the key relationship:

- If energy is transferred or transformed, then work has been done.
- Conversely, if no energy is transferred or transformed, no work has been done.

The teacher will present simple scenarios for students to consider (using chat or polling):

- A student holding a book steady at arm's length. Is work being done? (No, because the book's energy is not changing—no transfer is occurring).
- A student carrying a book horizontally across the room at constant speed. Is work being done on the book? (No, because the book's potential energy (height) and kinetic energy (speed) remain constant—no net energy transfer is happening to the book).
- A student lifting the same book upward. Is work being done? (Yes, because the book's potential energy is increasing—energy is being transferred from the student to the book).

The teacher will conclude this section with a concise summary:

- Energy is the capacity to do work (the "currency").
- Work is the transfer or transformation of energy (the "transaction").
- Wherever you see energy change form or move from one object to another, work is happening.

D. ELABORATE (20 minutes)

The teacher will have the students solve problems about kinetic energy, potential energy, and the conservation of energy.

Problem 1: A **2 kg** object is dropped from a height of **15 m**. a. Calculate the initial potential energy before the drop. b. Calculate the maximum kinetic energy of the object as it falls.

Problem 2: A pendulum bob with a mass of **0.25 kg** is displaced from equilibrium until it reaches a vertical height of **0.45 m** above its lowest point. Calculate the kinetic energy and the velocity of the bob when it is exactly **0.15 m** above the equilibrium position.

E. Evaluate (15 minutes)

The teacher begins the evaluation phase through Google forms. The google forms will contain 10 multiple choice items, each worth 1 point, and 2 word problems.

IV. Learning Assessment

The teacher will administer a short formative assessment to measure the learners' understanding of kinetic energy, potential energy, mechanical energy, conservation of

energy, and work. The evaluation will consist of three parts: Conceptual Understanding and problem solving.

1. Which of the following best describes kinetic energy?
A. Energy stored due to position
B. Energy of motion
C. Energy from electricity
D. Energy from heat

2. When a ball is at the highest point of its motion, it has:
A. Maximum kinetic energy
B. Minimum potential energy
C. Maximum potential energy
D. No mechanical energy

3. Mechanical energy is equal to:
A. KE – PE
B. KE × PE
C. KE + PE
D. KE ÷ PE

4. If an object is falling freely (no friction), its total mechanical energy:
A. Increases
B. Decreases
C. Remains constant
D. Becomes zero

5. Work is done on an object when:
A. The object is at rest
B. There is energy transfer
C. The object has mass
D. The object has height

6. Which of the following best defines energy?
A. The speed of an object
B. The ability to do work
C. The force applied to an object
D. The mass of an object

7. Kinetic energy depends on an object's:
A. Height and gravity
B. Mass and velocity

C. Temperature and mass
D. Height and time

8. The formula for kinetic energy is:
A. KE = mgh
B. KE = mv
C. KE = ½mv²
D. KE = mg

9. Gravitational potential energy depends on:
A. Speed and mass
B. Height and velocity
C. Mass, gravity, and height
D. Force and time

10. A ball at the highest point of a roller coaster track has:
A. Maximum kinetic energy
B. Maximum potential energy
C. No mechanical energy
D. Zero mass

Word Problem:

A **5 kg** marble is dropped from the edge of a big bowl. The bowl's edge is **3 m** higher than its lowest point in the middle. In the first snapshot of the system, the marble is traveling at **1.19 m/s**. In the second snapshot, the marble has rolled up **1 m** from the base.

a. Calculate the total mechanical energy of the system.

b. Calculate the kinetic energy of the marble in the first snapshot, then use the law of conservation of energy to determine its height at that exact moment.

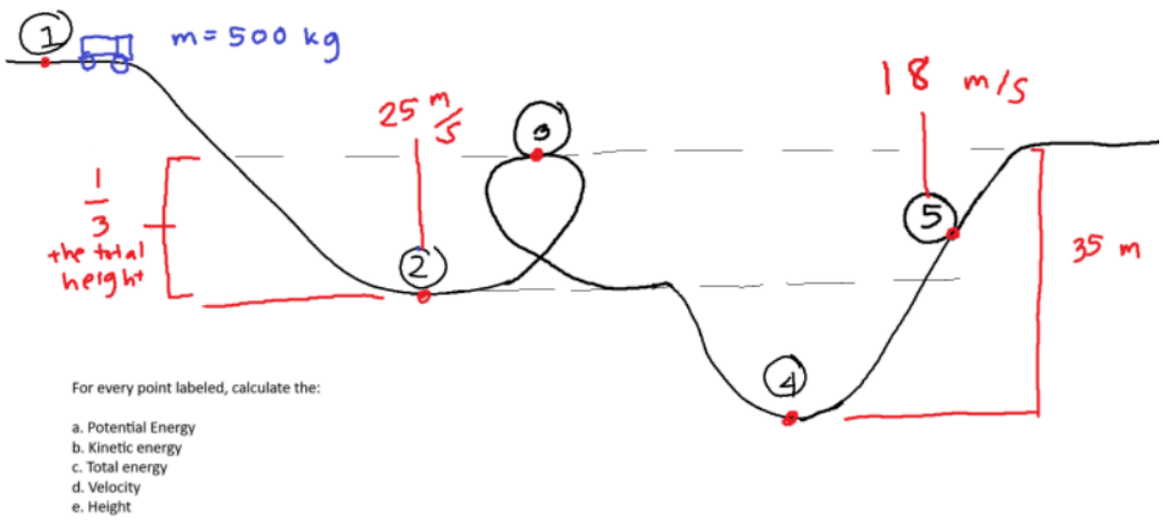
c. Calculate the potential energy of the marble in the second snapshot, then use the law of conservation of energy to determine its velocity at that exact moment.

V. Asynchronous Task (Enrichment or Agreement)

The teacher will provide the students an answer sheet for them to work on:

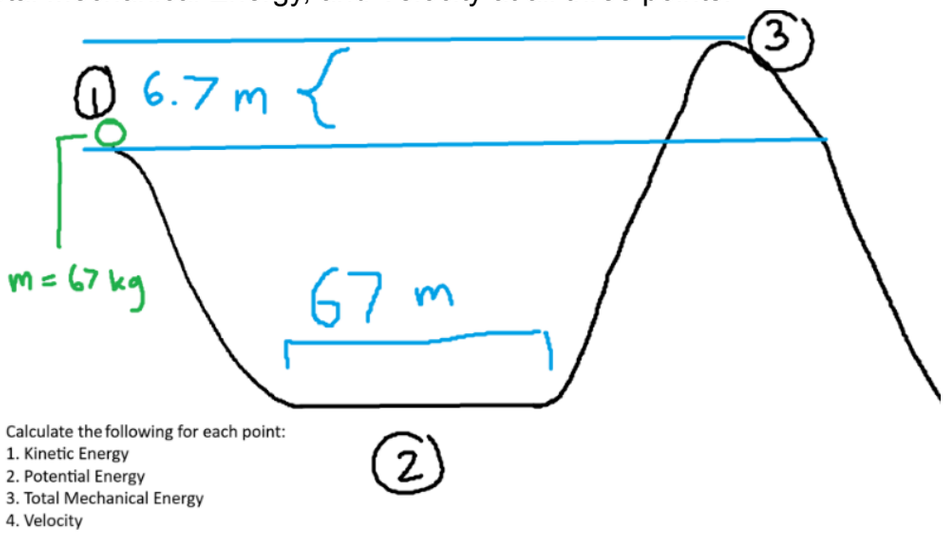
1. A **500 kg** roller coaster car is released from rest at Point 1, the highest elevation of a frictionless track. At Point 2, the car travels at **25 m/s**. The track elevation increases from Point 2 to a loop apex at Point 3, which aligns exactly with the final, flat plateau measuring **35 m** above the track's absolute baseline (Point 4). The vertical displacement between Point 2 and the **35 m** elevation

mark is geometrically defined as exactly one-third of the total track height. The car ascends the final ramp, passing Point 5 at a velocity of **18 m/s**. Use g = 9.8 m/s^2



2. (A) A **67 kg** mass is released from rest at Point 1 on a frictionless track. It descends to Point 2, a flat horizontal segment measuring exactly **67 m** in length. The mass takes precisely **5 s** to traverse this segment. After Point 2, the track ascends to a peak at Point 3. Geometric measurements confirm Point 3 is located precisely **6.7 m** above the initial elevation of Point 1. Point 2 serves as the absolute elevation baseline (h = **0 m**). Use g = **9.8 m/s²**.

   Determine if the mass can successfully reach Point 3. If it cannot, provide the mathematical proof. If it can, calculate the Kinetic Energy, Potential Energy, Total Mechanical Energy, and Velocity at all three points.



   (B) The scenario is reversed. The **67 kg** mass is relocated and released from rest exactly at Point 3, traveling backward along the track. Calculate the same

four values at each point with this new scenario in mind.